

Library Management System Project Report

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1 Introduction

This project is a simple command-line based Library Management System written in Java. It supports operations for both users and librarians, including borrowing and returning books, managing book inventories, and user registration.

2 System Overview

The system is designed with an object-oriented approach and consists of several classes, each representing a component of the library system.

3 Class Descriptions

3.1 Book

The **Book** class contains basic information about a book including its name, author, publication year, and ISBN. It provides getter and setter methods to access and modify these attributes.

3.2 User

The **User** class represents a library user. It includes a username, password, and a list of borrowed books. It allows users to rent and return books.

3.3 Librarian

The **Librarian** class holds credentials of library administrators who are authorized to manage the book collection.

3.4 Library

This is the core class of the system. It maintains lists of books, users, and librarians and provides functionality to:

- Add, remove, update, and search for books
- Add, remove, update, and search for users
- Add, remove, and search for librarians
- Track the quantity of each book

3.5 Main

The **Main** class runs the application. It includes the menu-driven interface where users and librarians can interact with the system.

4 Functionalities

4.1 User Operations

- Login and logout
- Register as a new user
- Borrow books
- Return books

4.2 Librarian Operations

- Login and logout
- Add new books
- Remove books
- Increase/decrease quantity of books

5 Sample Code Snippet

Here is an example of how the system allows a user to borrow a book:

```
if (!library.doesBookExist(name, author, year, isbn)) {  
    System.err.println("This book doesn't exist!");  
} else {  
    library.removeBook(name, author, year, isbn);  
}
```

6 Conclusion

The project demonstrates fundamental principles of object-oriented programming. Future enhancements may include persistent storage, a GUI, or user authentication with encryption.